

## HOERNLI, Switzerland

WMO: 066890

Lat: 47.3708N

Lon: 8.9417E

Elev: 1124

StdP: 88.53

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -11.8      | -9.7 | -18.2                         | 0.9 | -4.4 | -14.7 | 1.2 | -3.4 | 17.4                  | 5.2  | 15.2 | 4.0  | 1.8                   | 80   | 0.606 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 6.5                    | 25.2            | 17.0 | 23.6 | 16.3 | 22.1 | 15.5 | 18.2                | 23.4 | 17.1 | 22.1 | 16.3 | 20.9 | 2.1                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 16.2                            | 13.2 | 21.0 | 15.2 | 12.4 | 19.6 | 14.3 | 11.7 | 18.6 | 56.1          | 23.4 | 52.7 | 22.2 | 49.9 | 21.0 | 22.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 12.7              | 10.7 | 9.1 | -13.3                      | 27.3 | 2.8                | 1.7 | -15.3                                              | 28.6 | -16.9      | 29.6 | -18.5      | 30.5 | -20.6      | 31.8 |    |    |
|                   |      |     | -13.7                      | 20.3 | 2.9                | 1.3 | -15.8                                              | 21.3 | -17.5      | 22.1 | -19.1      | 22.8 | -21.3      | 23.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 7.2    | -0.8  | -0.7  | 2.5   | 6.2   | 10.3  | 13.9  | 15.7  | 15.5  | 11.5  | 8.3   | 3.3   | 0.4  |
|                                                                                  | DBStd       | 7.50   | 4.60  | 5.16  | 4.83  | 4.91  | 4.37  | 4.47  | 4.00  | 4.08  | 3.95  | 4.47  | 4.92  | 4.78 |
|                                                                                  | HDD10.0     | 1692   | 334   | 300   | 234   | 133   | 51    | 13    | 4     | 4     | 29    | 86    | 206   | 298  |
|                                                                                  | HDD18.3     | 4106   | 592   | 533   | 490   | 364   | 250   | 144   | 101   | 107   | 207   | 312   | 451   | 556  |
|                                                                                  | CDD10.0     | 681    | 0     | 0     | 2     | 19    | 61    | 132   | 181   | 175   | 74    | 33    | 5     | 0    |
|                                                                                  | CDD18.3     | 54     | 0     | 0     | 0     | 0     | 2     | 12    | 19    | 20    | 1     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 164    | 0     | 0     | 0     | 0     | 2     | 36    | 63    | 63    | 1     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 12     | 0     | 0     | 0     | 0     | 0     | 2     | 6     | 4     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.8    | 4.5   | 4.2   | 4.5   | 3.6   | 3.5   | 3.0   | 3.1   | 3.1   | 3.2   | 3.8   | 4.2   | 4.7  |
| Precipitation                                                                    | PrecAvg     | 1333   | 79    | 82    | 89    | 96    | 144   | 151   | 150   | 147   | 109   | 92    | 96    | 99   |
|                                                                                  | PrecMax     | 1693   | 190   | 210   | 242   | 192   | 293   | 243   | 283   | 260   | 229   | 172   | 213   | 219  |
|                                                                                  | PrecMin     | 1052   | 6     | 25    | 25    | 13    | 32    | 61    | 59    | 21    | 50    | 3     | 0     | 0    |
|                                                                                  | PrecStd     | 167    | 49    | 44    | 48    | 47    | 63    | 46    | 61    | 58    | 45    | 44    | 53    | 53   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 11.3  | 11.5  | 14.4  | 18.9  | 22.7  | 26.7  | 27.2  | 27.1  | 21.7  | 19.5  | 15.7  | 12.5 |
|                                                                                  |             | MCWB   | 4.0   | 4.0   | 7.4   | 10.3  | 13.9  | 18.1  | 18.1  | 17.3  | 15.5  | 11.1  | 8.1   | 5.2  |
|                                                                                  | 2%          | DB     | 8.6   | 9.4   | 12.5  | 17.0  | 20.6  | 24.2  | 25.1  | 25.1  | 19.9  | 17.3  | 13.8  | 10.0 |
|                                                                                  |             | MCWB   | 3.4   | 3.3   | 6.6   | 9.6   | 13.1  | 16.4  | 17.0  | 16.8  | 14.5  | 11.2  | 7.5   | 3.9  |
|                                                                                  | 5%          | DB     | 6.9   | 7.7   | 10.9  | 15.1  | 18.7  | 22.4  | 23.3  | 23.3  | 18.6  | 15.7  | 12.2  | 7.9  |
|                                                                                  |             | MCWB   | 2.8   | 2.6   | 5.7   | 8.6   | 12.1  | 15.6  | 16.3  | 16.4  | 13.8  | 10.7  | 6.6   | 3.2  |
|                                                                                  | 10%         | DB     | 5.3   | 6.0   | 9.3   | 13.3  | 16.9  | 20.8  | 21.7  | 21.6  | 17.2  | 14.3  | 10.2  | 6.5  |
|                                                                                  |             | MCWB   | 2.0   | 2.0   | 4.8   | 7.6   | 11.2  | 14.9  | 15.5  | 15.7  | 13.0  | 10.1  | 6.2   | 2.8  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 6.3   | 5.8   | 8.6   | 11.5  | 15.5  | 19.5  | 19.6  | 19.3  | 16.9  | 13.8  | 9.5   | 6.5  |
|                                                                                  |             | MCDB   | 7.5   | 8.7   | 12.4  | 17.4  | 21.0  | 24.9  | 25.5  | 24.3  | 20.3  | 16.6  | 13.2  | 8.7  |
|                                                                                  | 2%          | WB     | 4.7   | 4.6   | 7.5   | 10.3  | 14.0  | 17.6  | 18.0  | 18.1  | 15.5  | 12.5  | 8.5   | 5.4  |
|                                                                                  |             | MCDB   | 7.1   | 7.4   | 11.2  | 15.7  | 19.1  | 22.8  | 23.5  | 23.1  | 18.8  | 15.1  | 11.7  | 8.0  |
|                                                                                  | 5%          | WB     | 3.6   | 3.6   | 6.4   | 9.2   | 12.8  | 16.4  | 17.0  | 17.0  | 14.5  | 11.6  | 7.5   | 4.6  |
|                                                                                  |             | MCDB   | 6.1   | 6.5   | 9.8   | 14.1  | 17.4  | 21.2  | 22.1  | 21.8  | 17.6  | 14.3  | 10.6  | 6.9  |
|                                                                                  | 10%         | WB     | 2.5   | 2.6   | 5.4   | 8.2   | 11.7  | 15.3  | 16.1  | 16.1  | 13.5  | 10.7  | 6.6   | 3.6  |
|                                                                                  |             | MCDB   | 4.8   | 5.3   | 8.4   | 12.5  | 16.0  | 19.8  | 20.7  | 20.6  | 16.5  | 13.5  | 9.7   | 5.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.2   | 4.2   | 5.1   | 6.1   | 6.5   | 6.5   | 6.5   | 5.8   | 5.1   | 4.9   | 4.3   | 4.3  |
|                                                                                  |             | MCDBR  | 5.9   | 6.2   | 6.9   | 8.0   | 8.2   | 8.1   | 8.1   | 7.5   | 6.5   | 6.4   | 6.4   | 5.9  |
|                                                                                  |             | MCWBR  | 3.3   | 3.6   | 4.3   | 5.0   | 5.0   | 5.3   | 5.1   | 4.8   | 4.5   | 4.0   | 3.2   | 3.1  |
|                                                                                  | 5% WB       | MCDBR  | 5.1   | 5.6   | 6.4   | 7.7   | 7.9   | 7.8   | 7.7   | 7.0   | 6.0   | 5.5   | 5.9   | 5.2  |
|                                                                                  |             | MCWBR  | 3.7   | 3.8   | 4.4   | 4.9   | 5.1   | 5.3   | 5.2   | 4.8   | 4.6   | 3.9   | 3.7   | 3.6  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.260  | 0.281 | 0.312 | 0.344 | 0.357 | 0.363 | 0.365 | 0.358 | 0.334 | 0.310 | 0.283 | 0.256 |      |
|                                                                                  | taud        | 2.465  | 2.420 | 2.414 | 2.372 | 2.374 | 2.388 | 2.373 | 2.408 | 2.461 | 2.522 | 2.537 | 2.476 |      |
|                                                                                  | Ebn at Noon | 849    | 896   | 913   | 910   | 906   | 900   | 892   | 885   | 878   | 851   | 813   | 810   |      |
|                                                                                  | Edh at Noon | 63     | 82    | 97    | 112   | 116   | 116   | 116   | 107   | 93    | 75    | 60    | 56    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.20   | 2.05  | 3.26  | 4.34  | 4.97  | 5.72  | 5.54  | 4.83  | 3.63  | 2.30  | 1.29  | 0.98  |      |
|                                                                                  | RadStd      | 0.13   | 0.27  | 0.44  | 0.63  | 0.50  | 0.53  | 0.50  | 0.49  | 0.37  | 0.25  | 0.16  | 0.15  |      |

## Historical Trends

|      |                          | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|--------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                          |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only             | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (121 neighbors) | N/A   | N/A     | N/A    | +0.70   | +0.41 | +0.33 | N/A         | N/A     | +57     | +16     |

Nomenclature: See separate page